A review of Three Articles on Using Wi-Fi for Passive Motion Tracking and Gesture Recognition

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Location awareness and human gesture recognition are key enablers for a wide range of applications such as smart homes, virtual reality, augmented reality, security monitoring, and asset management. Traditional approaches track a user in an active manner via devices such as smartphones or wearable devices or sensors attached to users. While promising, these approaches pose inconvenience due to their respective drawbacks including leakage of privacy, requirement of on-body sensors and limits of sensing range, and thus are inapplicable in some scenarios such as security surveillance. Other approaches work passively with infrastructure installed in the area of interests, such as cameras and wireless sensors, which require the target to be in the direct line-of-sight (LOS) of the device. Because of such drawbacks and limitations, much effort was spent on developing devices which can perform passive tracking based on signals reflected off the target. Building a passive tracking system, however, is difficult because reflected signals are orders of magnitude weaker than directly received powers and are superimposed with reflections off other objects at the receiver. In this paper three papers are presented which introduce Widar[1]; a WiFi based passive motion tracking device which uses multiple links; its successor Widar2.0[2] which does the same but with only a single link; and finally Widar3.0[3] which extends both previous systems to be able to perform zero-effort cross-domain gesture recognition.

We begin our review with the original Widar. Widar builds on recent works on innovations in wireless communications which shed light on passive human sensing with Wi-Fi signals. In principle, these works exploit the phenomenon that human motions distort multipath profiles during signal propagation. These radio frequency (RF) based approaches are more attractive than previous solutions since they do not require the user to carry a device. They support omnidirectional coverage even in Non-Line-Of-Sight (NLOS) scenarios, and they preserve user privacy gracefully. Existing works, however, cannot track fine-grained human mobility information (including velocity, direction and location). Widar measures the instantaneous velocity, in order to track fine-grained continuous locations. Widar achieves these goals by deriving velocities and locations directly from the CSI of Wi-Fi signals. Channel State Information (CSI) in simple terms is all what happens in the medium in-between the emitter and receiver of Wi-Fi signals. The paper models it as a complex function H, with packet frequency rate and time as parameters. To eliminate unknown phase shifts which pollute the CSI, the paper uses the CSI power (i.e. the conjugate multiplication of CSI). Through Doppler Frequency Shifts (DFS), using the CSI the Paper describes a method for extracting the Path Length Change Rate (PLCR), i.e. the rate of change of each signal propagation path length, which coupled with some geometric analysis allows them to track a target's velocity. The basic geometric model places the

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[2] Qian, Kun, Chenshu Wu, Yi Zhang, Guidong Zhang, Zheng Yang, and Yunhao Liu. "Widar2. 0: Passive human tracking with a single Wi-Fi link." In *Proceedings of the 16th Annual International Conference on Mobile Systems, Applications, and Services*, pp. 350-361. 2018.

[3] Zheng, Yue, Yi Zhang, Kun Qian, Guidong Zhang, Yunhao Liu, Chenshu Wu, and Zheng Yang. "Zero-effort cross-domain gesture recognition with Wi-Fi." In *Proceedings of the 17th Annual International Conference on Mobile Systems, Applications, and Services*, pp. 313-325. 2019.

emitter and receiver in foci points of ellipses, and the target as a point on one of these ellipses. The paper then describes a method for estimating on which of these ellipses the target is located, and presents a way to calculate the radial velocity of the target based on PLCR estimates. Due to the conjugate multiplication of CSI in one of the previous steps, the sign of PLCR is unknown. Hence with 3 or more links, we will always have 2 possible velocities for the target. The issue can be solved as long as the sign of PLCR of any link is known. Widar does this by using a system called WirDir[4]. To detect movement Widar monitors its spectrograms. If it finds a large enough change in variance in of the power distribution of a frequency spectrum in the last two seconds before the user starts moving it will start tracking human movement. To estimate the initial location Widar first searches through the whole tracking area at a coarse resolution, identifying a location that yields the least fitting error of the trace, as the initial location. Then, Widar searches through the narrowed area around the initial location at a fine-grained resolution to refine the initial location accuracy. To avoid error accumulation Widar occasionally re-initializes the tracking process. To evaluate Widar the researchers used common-off-the-shelf (COTS) mini desktops, creating 6 links in total. The tracking area was 4x4 meters. They used an external camera to track the "ground truth" position and they set up the links in 2 different ways. Overall Widar outperformed single-link systems using PLCR as their only measurement and WirDir in terms of velocity direction estimation. With the starting location, across all paths Widar achieved a tracking error of 25cm. Without the starting location accuracy was around 38cm. Accuracy, however, degraded when subject moved further away from the links (up to 78cm). The paper also found greater error accumulation with longer walking distances, and a slight impact on performance with decreased packet rates. The paper did not find that Widar was impacted by walking direction, link arrangements and human subject diversity.

The main disadvantage of Widar is its requirement of multiple links which makes it less favorable for practical situations, especially in smart home where people would like to only deploy a few devices for sensing. Widar2.0 enables passive human localization and tracking using a single link by leveraging multi-dimensional signal parameters such as Angle-of-Arrival (AoA), Time-of-Flight (TOF) and Doppler Phase Shifts (DFS). Widar2.0 consists of two key modules: the CSI-Motion module and the Motion-Tracking module. The CSI-Motion module conducts joint multiple parameters estimation of multipath signals from noisy CSIs. Upon receiving CSI measurements, Widar2.0 first cleans them using conjugate multiplications of CSIs just like Widar. Thereafter, Widar2.0 applies a parameter estimation algorithm to obtain multidimensional parameters (ToF, AoA and DFS) of multipath signals. The problem of joint multiple parameter estimation was formulated as a maximum likelihood estimation (MLE) problem which the paper solves by applying the Space Alternating Generalized Expectation Maximization (SAGE)[5] algorithm. The Motion-Tracking module calculates the target locations from multiple signal parameters estimates. Then, the ToF and DFS are fused through a Kalman Smoother to refine distance estimates. The localization model of Widar2.0 is very similar to Widar, except that now, becuase Widar2.0 has AoA and ToF in addition to DFS information, using geometry it can produce a unique solution for the location of the subject being tracked with only a single link. To evaluate Widar2.0 the researchers used a pair of OTS laptops, with a single transmitter antenna and three receiver antennas. The transmitter and receiver were placed

[4] Wu, Dan, Daqing Zhang, Chenren Xu, Yasha Wang, and Hao Wang. "WiDir: walking direction estimation using wireless signals." In *Proceedings of the 2016 ACM international joint conference on pervasive and ubiquitous computing*, pp. 351-362. 2016.
[5] Fessler, Jeffrey A., and Alfred O. Hero. "Space-alternating generalized expectation-maximization algorithm." *IEEE Transactions on signal processing* 42, no. 10 (1994): 2664-2677.

in two adjacent corners of the tracking areas. They also tested Widar2.0 in three different environments: 1) an empty classroom, 2) a small office with furniture 3) a narrow corridor. In the classroom they also used 2 receivers placed orthogonally to test improvements in accuracy in using 2 links vs. 1 link. They used 6 volunteers who walked in different trajectories. The average localization error of Widar2.0 using a single link was 0.75m. Using two links Widar 2.0 achieved an error of 0.63m. Widar2.0 also outperformed Dynamic Music[6] which uses only AoA. Widar2.0 did slightly worse than IndoTrack[7] (localization error of 0.48m). However, Widar2.0 was much better at avoiding error accumulation with increased walking distance. Widar 2.0 was also impacted by walking direction. Using a single link there was a high degradation in localization error when subjects walked parallel with the link. Two links proved much more accurate at addressing different walking directions. Location error was stable (0.6m) for all velocities, yet gradually decreased as velocities increased. Widar2.0 was robust to multipath effect in indoor environments. However, performance degradation was observed in larger tracking areas. Distance between the transmitter and receiver (i.e. link length) also impacted the system performance. Smaller link lengths lead to better accuracies. Just like Widar, Widar2.0 was not significantly impacted by packet rate. The researchers also conducted preliminary experiments where two people walked in the monitoring area. The results show that while Widar2.0 accurately estimates DFS and recognizes two people, it fails to yield accurate AoA and ToF estimates, and further track multiple people.

Unlike Widar and Widar2.0, Widar3.0's main focus is gesture recognition. Due to lack of spatial resolution, wireless signals and their features are highly specific to the environment where the gesture is performed, and the location and orientation of the performer. The paper terms these factors irrelevant to gestures as the "domain". The traditional approaches for gesture recognition used classifiers trained with primitive signal features in one domain, and had to do extra training for each new domain that was introduced. Because the location and orientation of a person are continuous, there exist an infinite number of domains, making cross-domain training approaches practically prohibitive. Widar3.0 develops a one-fits-all model which enables cross-domain gesture recognition. Widar3.0 consists of two main modules: 1) the BVP generation module and 2) the gesture recognition module. Body-Coordinate Velocity Profile (BVP) describes power distribution over different velocities. Each type of gestures has its unique velocity profile in the body coordinate system (e.g., the coordinates where the orientation of the person is the positive x axis) no matter in which domain the gesture is performed. Widar3.0 extracts all its information from CSI. Widar3.0 uses the same process as Widar2.0 for extracting the DFS profile, based on which they derive a domain-independent BVP. The orientation and location information of the person is estimated via previously developed systems such as LiFS[8] IndoTrack and Widar2.0. When a person performs a gesture their body parts move at different velocities. As a result, signals reflected by these body parts experience various DFS, which are superimposed at the receiver and form the corresponding DFS profile. The DFS profile contains the information of the gesture, but it is highly coupled with the domain. Knowing the persons' position and direction

[6] Li, Xiang, Shengjie Li, Daqing Zhang, Jie Xiong, Yasha Wang, and Hong Mei. "Dynamic-music: accurate device-free indoor localization." In *Proceedings of the 2016 ACM International Joint Conference on Pervasive and Ubiquitous Computing*, pp. 196-207. 2016.

[7] Li, Xiang, Daqing Zhang, Qin Lv, Jie Xiong, Shengjie Li, Yue Zhang, and Hong Mei. "IndoTrack: Device-free indoor human tracking with commodity Wi-Fi." *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 1, no. 3 (2017): 1-22.

[8] Wang, Ju, Hongbo Jiang, Jie Xiong, Kyle Jamieson, Xiaojiang Chen, Dingyi Fang, and Binbin Xie. "LiFS: Low human-effort, device-free localization with fine-grained subcarrier information." In *Proceedings of the 22nd Annual International Conference on Mobile Computing and Networking*, pp. 243-256. 2016.

allows Widar3.0 to decouple this information and create a unique BVP. To create an un-ambiguous BVP one must have at least 3 links. More links can be used to avoid edge-case ambiguities in estimating the BVP. Once BVP is obtained, the gesture recognition module then uses a Deep Neural Network (DNN) to mine spatial-temporal characteristics of the BVP series. To evaluate Widar3.0, the researchers used OTS transmitter (1 emitter and at least 3 receivers). They tested their system in 3 environments, similar to those of Widar2.0. The sensing area was a 2x2 meter square, and they assumed only 1 person was in the sensing area. 1 transmitter and 2 receivers were placed in corners of sensing area other receivers were placed randomly. All transcievers were placed at height of 110cm. They used 16 volunteers; each volunteer performed gestures in 5 locations and 5 orientations in each sensing area. They used 2 types of gestures: 1) common gestures (such as "push-pull") 2) complex gestures (such as drawing digits). Overall accuracy was 92.7% with 90 and 10 percentage data collected in Room 1 used for training and testing, respectively. Subjects were then instructed to perform a random gesture, which made accuracy drop to 90.1%. Widar3.0 can differentiate the unknown class with an accuracy of 87.1%. Researchers also found that accuracy degrades when gesture is directly in front of the user torso and is hence blocked (such as the push-pull gesture). Widar 3.0 is also designed to sense horizontally directed gestures so when a gesture has vertical components such as "drawing a circle", accuracy drops. Average accuracy for all locations (within the same room) that were not involved in testing were above 85%. Best accuracy was at the center of the room and the worst when subject was furthest away from the sensors. Widar3.0 was also robust to switching rooms, with an accuracy of 87%. The system was also robust to test subject diversity, reaching an accuracy of 85% across 7 subjects. The researchers also found increased accuracy with an increased number of links. For 3-6 links accuracy ranged from 80 to 90%. With 2 links accuracy dropped significantly to 70%. The accuracy remains over 90% when location error is within 40cm, but then drops. Accuracy gradually drops with higher orientation errors. The average gesture recognition accuracy drops from 89% to 74% when the number of people used for training drops from 7 to 1. Like its predecessors, Widar3.0 was not impacted by the transmission rates of the packets.

All three papers follow a very similar structure where they introduce, on a high level, how the new system works, after which, they go into low level details of how each part of their respective system was implemented. Finally, through experiments and their results they evaluate their systems' performance. All three systems use COTS devices and derive their parameters from CSI, which was cleaned using CSI conjugate multiplication. The main conceptual difference between Widar and Widar2.0 is that Widar estimates location by leveraging knowledge of the same parameter (DFS) across multiple links, whereas Widar2.0 estimates location by leveraging knowledge of multiple parameters (DFS, AoA and ToF) from a single link. To evaluate the robustness of Widar2.0 the researchers used more extensive experiments to test the impact of a wider variety of parameters. Even though Widar achieved smaller location errors than Widar2.0, both systems had better or comparable accuracy to other state-of-the-art technologies. The third paper proposes Widar3.0, which uses these motion and location tracking systems, and builds upon them to achieve cross-domain gesture recognition. In its evaluation phase it borrows aspects from Widar2.0; namely using multiple environments, but obviously testing the systems' ability to recognize gestures, rather than its ability to track the subject's position. Just like its predecessors, Widar3.0 achieved better or comparable performance to other state-of-the-art systems, on average keeping its accuracy at a impressive 85% across varying parameter values.

References:

[1] Qian, Kun, Chenshu Wu, Zheng Yang, Yunhao Liu, and Kyle Jamieson. "Widar: Decimeter-level passive tracking via velocity monitoring with commodity Wi-Fi." In Proceedings of the 18th ACM International Symposium on Mobile Ad Hoc Networking and Computing, pp. 1-10. 2017.

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